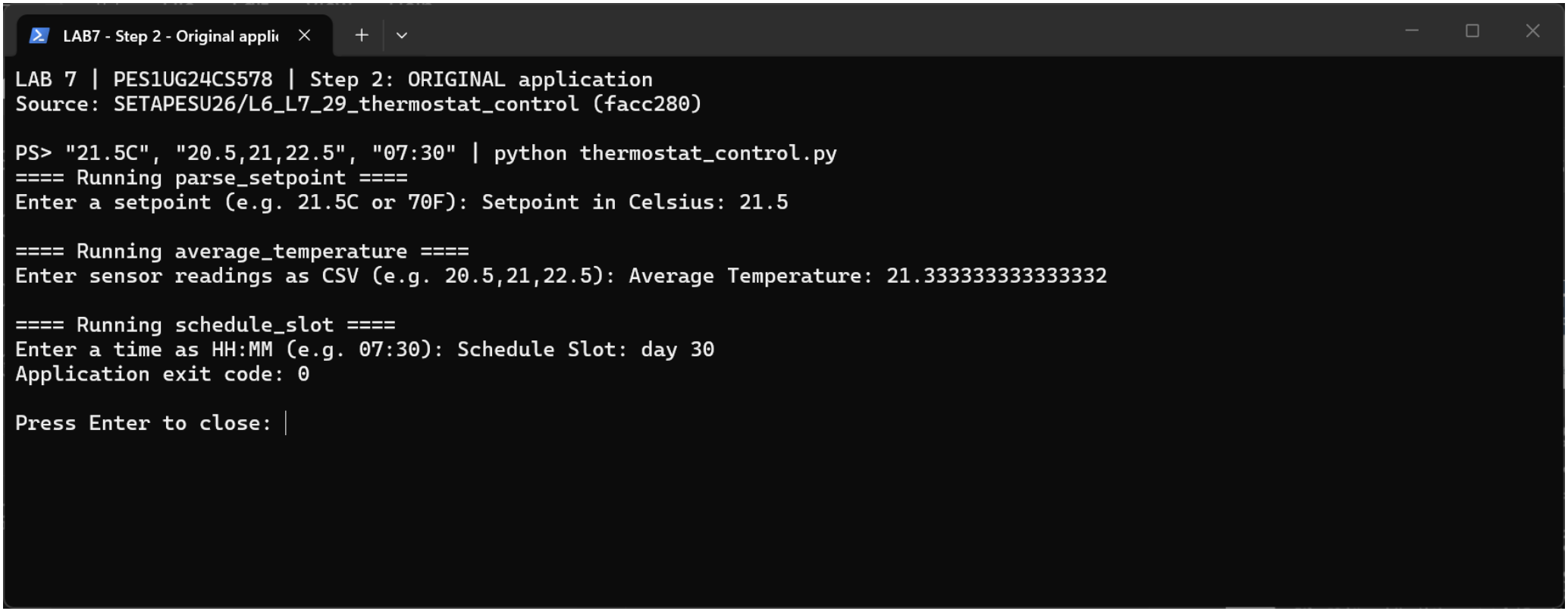
Lab 7: Fuzz Testing

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Step 2: Run the original application

Original program with valid sample inputs: 21.5C, readings 20.5/21/22.5, and time 07:30.

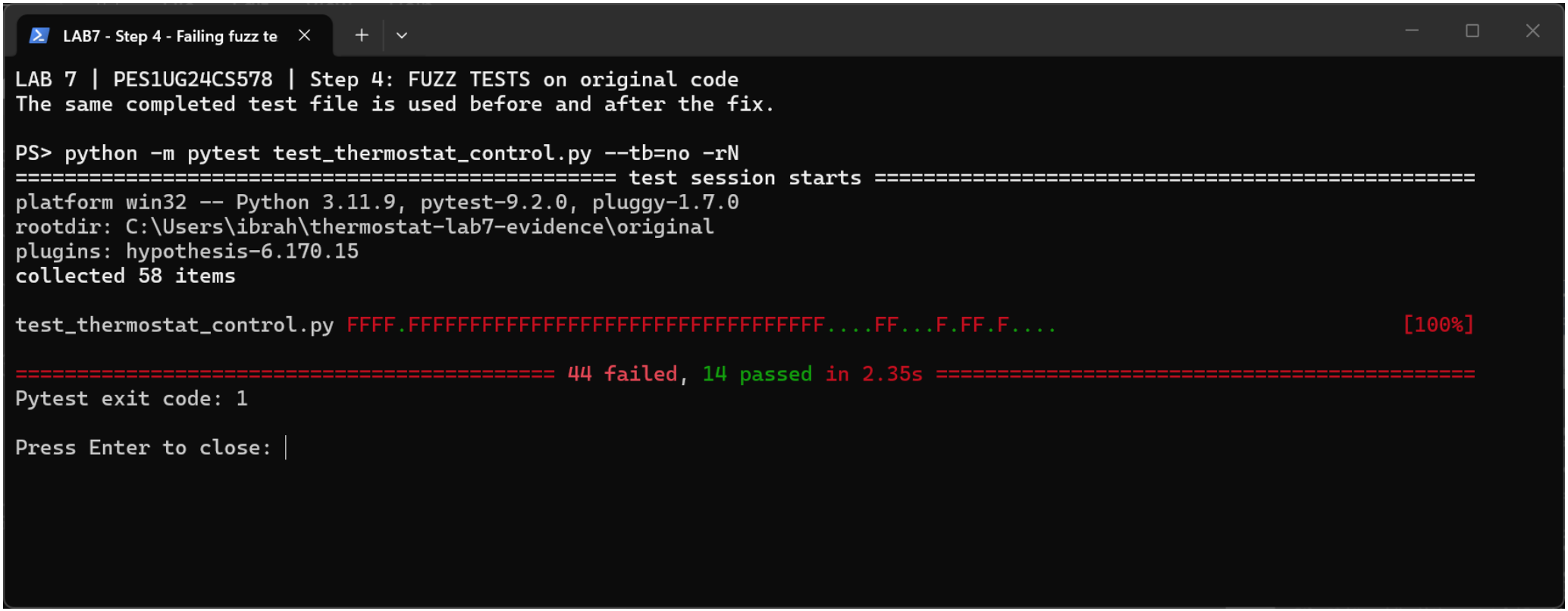


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Step 4: Run the completed fuzz tests on original code

Before fixing the code: 44 tests fail and 14 pass. These are the expected failures for Step 4.

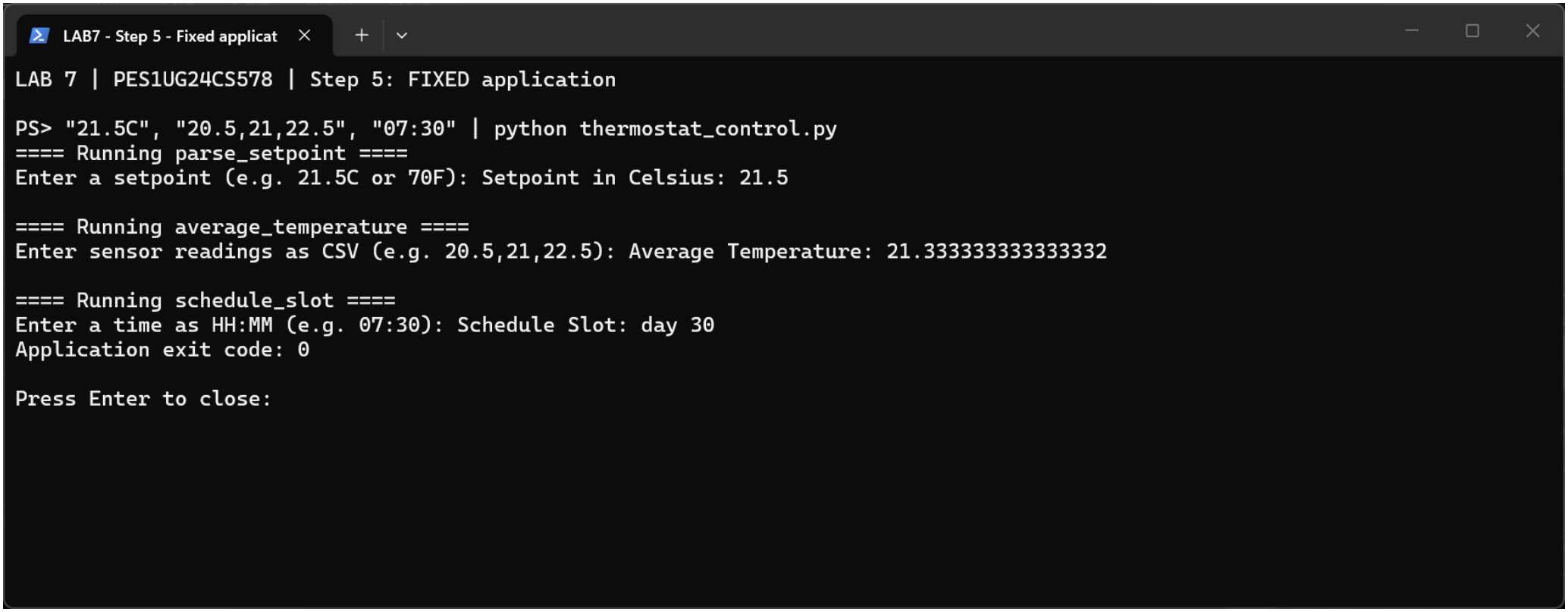


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Step 5: Run the fixed application

The fixed program runs successfully with the same sample inputs and exits with code 0.

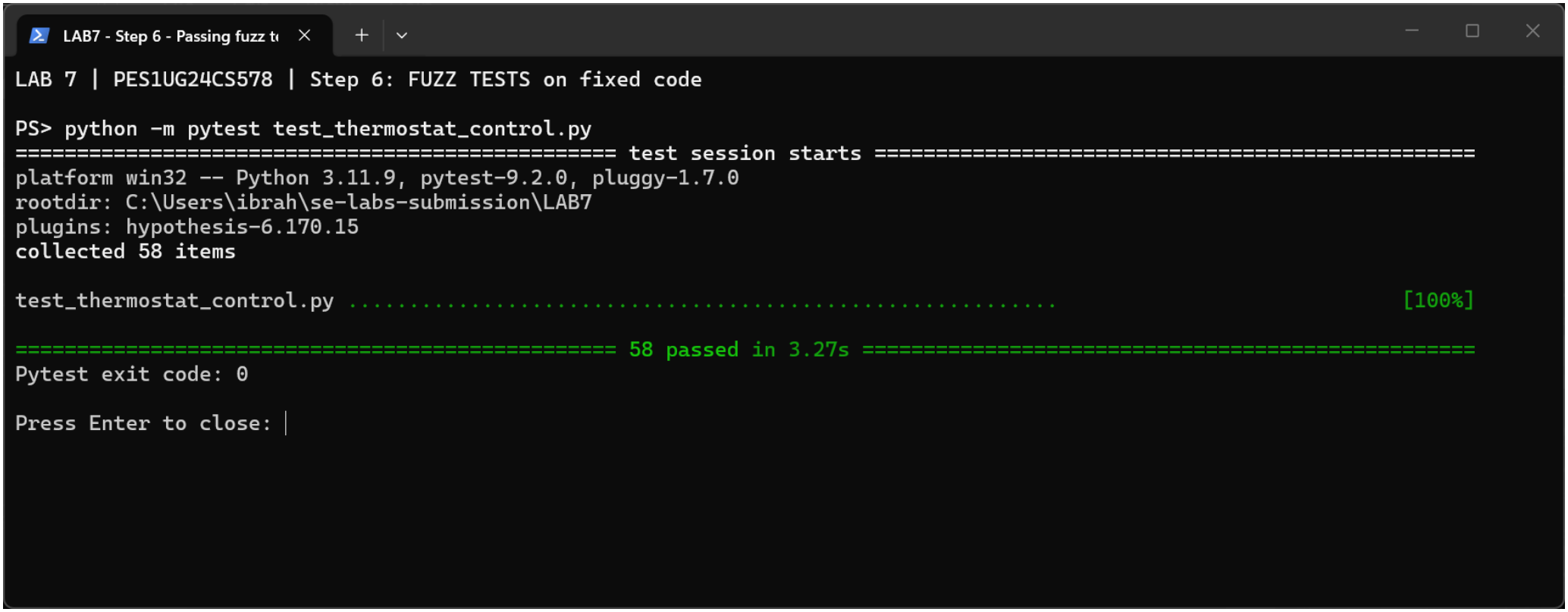


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Step 6: Re-run the fuzz tests on fixed code

All 58 tests pass. The test file is identical in the before and after runs; pytest exits with code 0.



Test design and reproduction

8 Hypothesis properties (up to 500 examples each) plus 50 parametrized regression cases.

Install: python -m pip install pytest hypothesis

Run inside LAB7: python -m pytest test_thermostat_control.py